

To : P.T. Honda Prospect Motor

Report No. : APTX26124
 Issued date : 21 May 2026
 NSK Asia Pacific Technology
 Centre (Thailand) Co., Ltd.
 Approved by : T. Ikeda
 Checked by : E. Watanabe
 : C. Pastraporn
 Prepared by : T. Theera

Investigation Report of Returned Front Wheel Bearing from Indonesia

1. Introduction

These hub bearings were returned from Indonesian market due to abnormal noise. NSK-APTC has performed an investigation to find out the cause of trouble.

2. Investigation bearings

Country : Indonesia
 Customer part No. : 44300-T9C -T511-M1
 NSK part No. : 38BWD27R
 Filling grease : JX Nippon Oil & Energy Corporation, Pyronoc Universal N6C
 Application : Front wheel hub unit bearing
 Quantity : 4 pcs.

Vehicle information

Brg. No.	Frame No.	Engine No.	Bearing production date	Delivery date	Repaired date	Mileage (km)	Dealer (code)
1L [#]	MHRDD4730PJ301152	L15Z17381152	27/01/2018	18/12/2023	09/09/2025	61,483	Honda Dewata Motor (0411)
3L [#]	MHRDD4730PJ301221	L15Z17381221	12/11/2019	11/01/2024	06/02/2026	80,469	Honda Dewata Motor (0411)
4R [#]	MHRDD4730PJ301217	L15Z17381217	01/02/2018	11/01/2024	05/03/2026	86,338	Honda Dewata Motor (0411)
5R [#]	MHRDD4730PJ301185	L15Z17381185	12/11/2019	18/12/2023	31/03/2026	87,173	Honda Dewata Motor (0411)

Remark : L = Left wheel side, R = Right wheel side.

: # This bearing might not be returned from this vehicle due to bearing production date is too difference from car delivery date.

3. Investigation items

1. Dimension and characteristic check : Page 2
2. Condition as received and grease condition : Page 3 - 6
3. Bearing raceways and parts condition : Page 3 - 6
4. Roundness : Page 7
5. Grease spectrum (FTIR) : Page 8

4. Investigation result

Group	Bearing No.	Bearing condition	Possible cause	Quantity
D	1L, 3L, 4R and 5R	Flaking was found on raceway. Also moisture content in grease was high.	Water penetration	4

Remark : L = Left wheel side, R = Right wheel side.

Result of investigation and analysis of each category is shown in following pages 2 - 8.

5. Summary

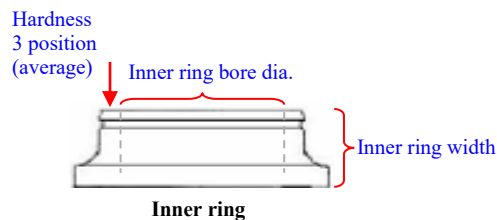
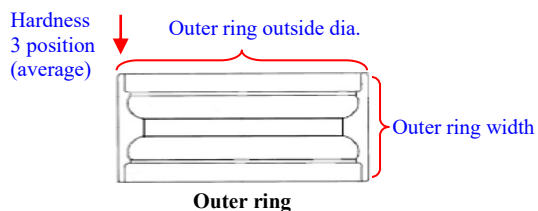
- Group D, 4 bearings : Bearing No. 1L, 3L, 4R and 5R.
 : Flaking was found on raceway. Hence the vehicle noise was caused by this failure. Also high moisture content in grease was found.
 This type of flaking is typically generated by water penetration.

Bearing inspection data

Item		Spec. of new bearing	Bearing No.			
			1L	3L	4R	5R
Bearing and parts condition						
Condition as received [Assembled condition]		-	- Bearing was returned in complete condition. - Outboard seal out of position from dismounting process was found since received. - Rough hand rotating feeling was found.	- Bearing was returned in complete condition. - Rough hand rotating feeling was found.	- Bearing was returned in complete condition. - Rough hand rotating feeling was found.	- Bearing was returned in complete condition. - Outboard seal out of position from dismounting process was found since received. - Rough hand rotating feeling was found.
Bearing internal condition after disassembled	Inboard	-	Grease color had changed.	Grease color had changed.	Grease color had changed.	Grease color had changed.
	Outboard	-	Grease color had changed.	Grease color had changed.	Grease color had changed.	Grease color had changed.
Outer ring raceway condition	Inboard	-	No abnormality	Indentation	Flaking	Indentation
	Outboard	-	Indentation	No abnormality	Flaking	No abnormality
Inner ring raceway condition	Inboard	-	No abnormality	Flaking	Flaking	Flaking
	Outboard	-	Flaking	No abnormality	Flaking	No abnormality
Balls condition	Inboard	-	No abnormality	Indentation	Indentation	Indentation
	Outboard	-	Indentation	No abnormality	Indentation	No abnormality
Sound pressure(dB) After wash @ 1500 rpm	Inboard	-	61	77	81	81
	Outboard	-	83	64	86	60
Bearing Dimension						
Outer ring outside diameter (mm)		Ø74 -0.013 ~ 0	-0.004	-0.007	-0.008	-0.005
Inner ring bore diameter (mm)	Inboard	Ø 38 -0.032 ~ -0.020	-0.021	-0.022	-0.023	-0.021
	Outboard		-0.015 *1	-0.023	-0.021	-0.024
Width outer ring (mm)		40 -0.060 ~ 0	-0.028	-0.038	-0.027	-0.023
Width inner ring (mm)	Inboard	20 -0.060 ~ 0	-0.045	-0.026	-0.026	-0.019
	Outboard		-0.037	-0.027	-0.024	-0.031
Axial clearance (μm)		65 ~ 85	_ *2	_ *2	_ *2	_ *2
Grease Condition						
Grease remain (%)		30% - 40%	32.9	33.3	35.0	35.2
Grease spectrum [FTIR]		-	No abnormality	No abnormality	No abnormality	No abnormality
Moisture content in grease (%wt)	Inboard	-	0.03	0.12	0.07	0.08
	Outboard	-	0.49	0.04	0.53	0.19
Hardness (HRC)						
Outer ring		HRC 60 ~ 64	63.1	62.9	62.8	62.9
Inner ring [Inboard / Outboard]		HRC 60 ~ 64	62.8 / 63.1	62.5 / 62.5	62.6 / 62.5	62.6 / 61.9
Ball [Inboard / Outboard]		HRC 61 ~ 66	64.2 / 65.4	64.2 / 64.3	64.5 / 64.7	64.0 / 64.6

Remark : *1 Over specification was probably caused by usage.

: *2 Cannot be measured due to flaking.



Photos of bearing No. 1L



Condition as received : Inboard
Bearing was returned in complete condition.
Smooth hand rotating feeling was found.



Condition as received : Outboard
Bearing was returned in complete condition.
Seal out of position from dismounting process was found since received.
Rough hand rotating feeling was found.



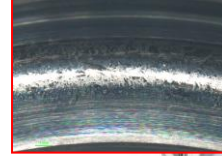
Internal condition : Inboard
Grease color had changed but base oil remained enough.



Internal condition : Outboard
Grease color had changed but base oil remained enough.



Outer ring condition : Inboard
No abnormality was found.



Outer ring condition : Outboard
Indentation was found on raceway.



Inner ring condition : Inboard
No abnormality was found.



Inner ring condition : Outboard
Flaking was found on raceway.



Balls and cage condition : Inboard
No abnormality was found.



Balls and cage condition : Outboard
Indentation was found on balls surface.
Breakage was found on cage.



Seal and encoder condition : Inboard
No abnormality was found on seal lip and encoder.
(Seal deformation from dismounting process was found.)



Seal condition : Outboard
No abnormality was found on seal lip.
(Seal deformation from dismounting process was found.)

Photos of bearing No. 3L



Condition as received : Inboard
Bearing was returned in complete condition.
Rough hand rotating feeling was found.



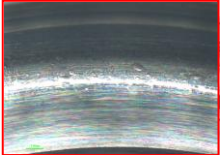
Condition as received : Outboard
Bearing was returned in complete condition.
Smooth hand rotating feeling was found.



Internal condition : Inboard
Grease color had changed but base oil remained enough.



Internal condition : Outboard
Grease color had changed but base oil remained enough.



Outer ring condition : Inboard
Indentation was found on raceway.



Outer ring condition : Outboard
No abnormality was found.



Inner ring condition : Inboard
Flaking was found on raceway.



Inner ring condition : Outboard
No abnormality was found.



Balls and cage condition : Inboard
Indentation was found on balls surface.
Breakage was found on cage.



Balls and cage condition : Outboard
No abnormality was found.



Seal and encoder condition : Inboard
No abnormality was found on seal lip and encoder.
(Seal deformation from dismounting process was found.)



Seal condition : Outboard
No abnormality was found on seal lip.
(Seal deformation from dismounting process was found.)

Photos of bearing No. 4R



Condition as received : Inboard
Bearing was returned in complete condition.
Rough hand rotating feeling was found.



Condition as received : Outboard
Bearing was returned in complete condition.
Rough hand rotating feeling was found.



Internal condition : Inboard
Grease color had changed but base oil remained enough.



Internal condition : Outboard
Grease color had changed but base oil remained enough.



Outer ring condition : Inboard
Flaking was found on raceway.



Outer ring condition : Outboard
Flaking was found on raceway.



Inner ring condition : Inboard
Flaking was found on raceway.



Inner ring condition : Outboard
Flaking was found on raceway.



Balls and cage condition : Inboard
Indentation was found on balls surface.
No abnormality was found on cage.



Balls and cage condition : Outboard
Indentation was found on balls surface.
No abnormality was found on cage.



Seal and encoder condition : Inboard
No abnormality was found on seal lip and encoder.
(Seal deformation from dismounting process was found.)



Seal condition : Outboard
No abnormality was found on seal lip.
(Seal deformation from dismounting process was found.)

Photos of bearing No. 5R



Condition as received : Inboard
Bearing was returned in complete condition.
Rough hand rotating feeling was found.



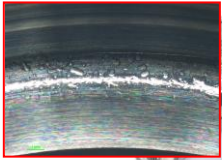
Condition as received : Outboard
Bearing was returned in complete condition.
Seal out of position from dismounting process was found since received.
Smooth hand rotating feeling was found.



Internal condition : Inboard
Grease color had changed but base oil remained enough.



Internal condition : Outboard
Grease color had changed but base oil remained enough.



Outer ring condition : Inboard
Indentation was found on raceway.



Outer ring condition : Outboard
No abnormality was found.



Inner ring condition : Inboard
Flaking was found on raceway.



Inner ring condition : Outboard
No abnormality was found.



Balls and cage condition : Inboard
Indentation was found on balls surface.
Breakage was found on cage.



Balls and cage condition : Outboard
No abnormality was found.

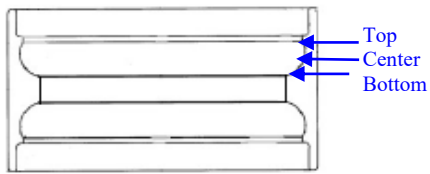


Seal and encoder condition : Inboard
No abnormality was found on seal lip and encoder.
(Seal deformation from dismounting process was found.)

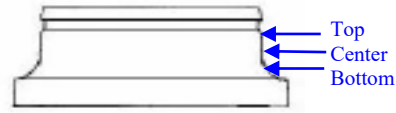


Seal condition : Outboard
No abnormality was found on seal lip.
(Seal deformation from dismounting process was found.)

Roundness of bearing



Outer ring

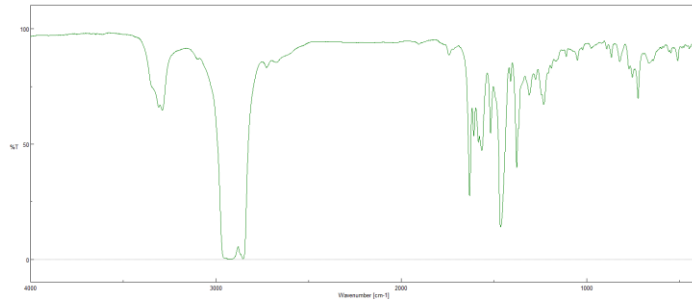


Inner ring

Bearing No. 1L			
Roundness: 1L A IB: 1.624µm Outer ring inboard side Roundness : 1.0 µm	Roundness: 1L A OB: 2.410µm Outer ring outboard side Roundness : 2.4 µm	Roundness: 1L B IB: 0.480µm Inner ring inboard side Roundness : 0.5 µm	Cannot be measured due to flaking. Inner ring outboard side Roundness : - µm
Bearing No. 3L			
Roundness: 3L A IB: 1.920µm Outer ring inboard side Roundness : 1.9 µm	Roundness: 3L A OB: 1.385µm Outer ring outboard side Roundness : 1.4 µm	Cannot be measured due to flaking. Inner ring inboard side Roundness : - µm	Roundness: 3L B OB: 0.361µm Inner ring outboard side Roundness : 0.4 µm
Bearing No. 4R			
Cannot be measured due to flaking. Outer ring inboard side Roundness : - µm	Cannot be measured due to flaking. Outer ring outboard side Roundness : - µm	Cannot be measured due to flaking. Inner ring inboard side Roundness : - µm	Cannot be measured due to flaking. Inner ring outboard side Roundness : - µm
Bearing No. 5R			
Roundness: 5R A IB: 1.529µm Outer ring inboard side Roundness : 1.5 µm	Roundness: 5R A OB: 1.037µm Outer ring outboard side Roundness : 1.0 µm	Cannot be measured due to flaking. Inner ring inboard side Roundness : - µm	Roundness: 5R B OB: 0.632µm Inner ring outboard side Roundness : 0.6 µm

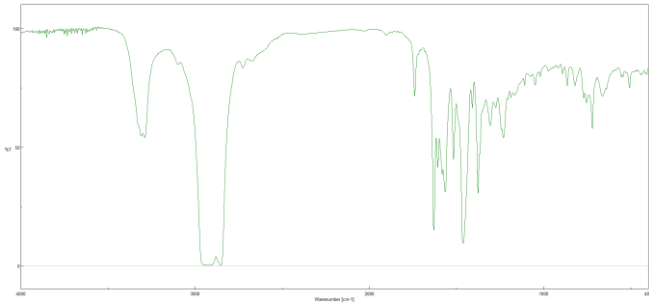
- Indentation was found on bearing No. 1L, 3L and 5R.

Grease spectrum result [FTIR Analysis]

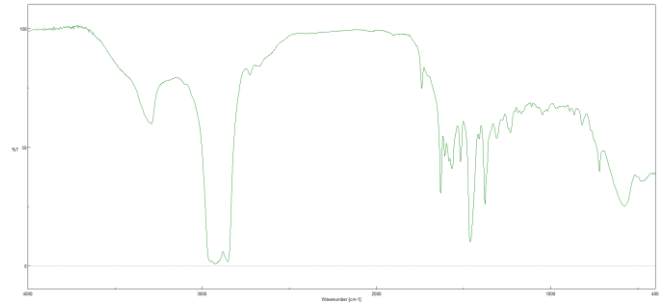


Spectrum of new JX Nippon Oil & Energy, Pyronoc Universal N6C grease

Bearing No. 1L

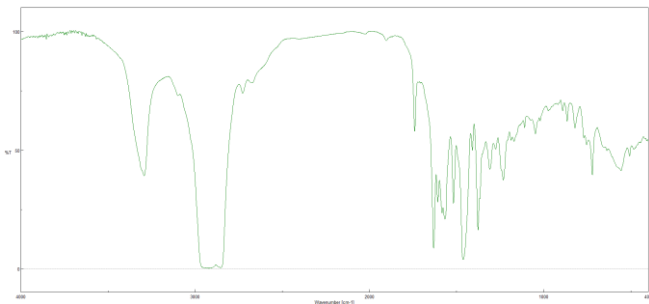


Spectrum of grease sample from inboard side.

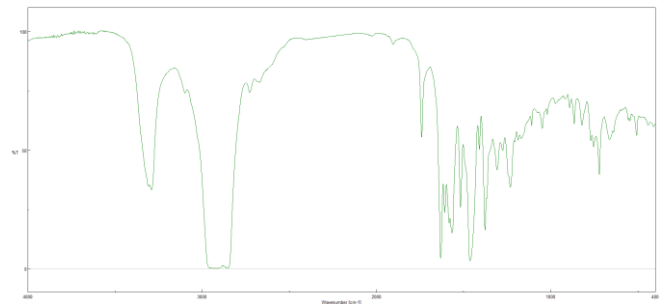


Spectrum of grease sample from outboard side.

Bearing No. 3L

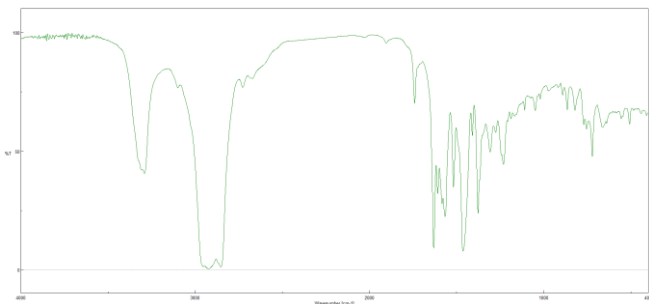


Spectrum of grease sample from inboard side.

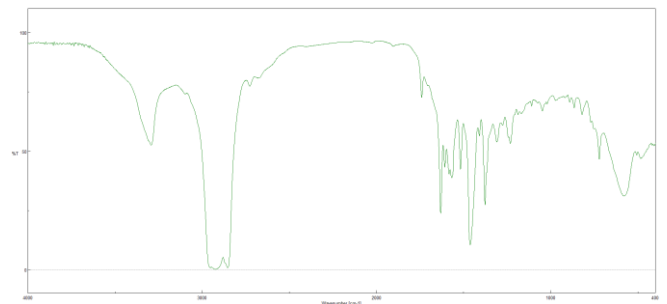


Spectrum of grease sample from outboard side.

Bearing No. 4R

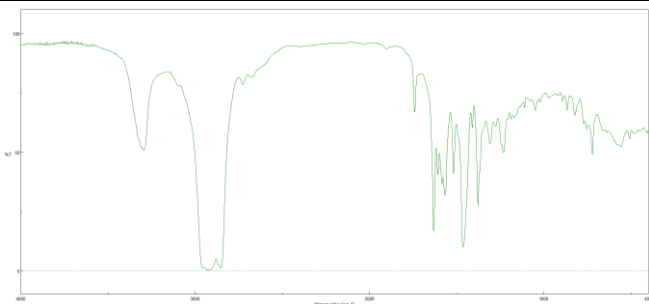


Spectrum of grease sample from inboard side.

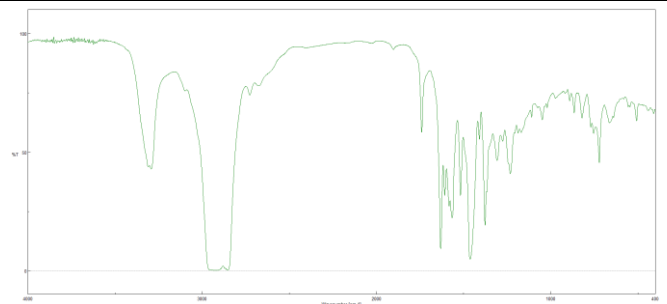


Spectrum of grease sample from outboard side.

Bearing No. 5R



Spectrum of grease sample from inboard side.



Spectrum of grease sample from outboard side.

- No abnormality was found.